Multimodal Video Summarization via Supervised Learning: Multimodal BiLSTM with Temporal Attention

and Evaluation of the VideoIntel Content Analysis Application

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Abstract

This report presents a supervised video summarization method evaluated on the SumMe and TVSum benchmarks. Visual features are extracted with a ResNet-50 backbone, while semantic speech features are obtained from Whisper automatic speech recognition and SentenceBERT sentence encodings. A two-layer Bidirectional LSTM (BiLSTM) combined with a Temporal Attention mechanism produces per-frame importance scores. Inference follows a fixed summary ratio of r=0.35 with Gaussian smoothing, shot segmentation, and boundary expansion to assemble the final dynamic summary. A complementary evaluation uses the VideoIntel application—built on CLIP, Whisper, and a local LLM—to measure the retention of search and question-answering capabilities when the original video is replaced by its summary. All content is self-contained; no external source files are required.

Keywords. video summarization, supervised learning, BiLSTM, temporal attention, multimodal, SumMe, TVSum, CLIP, Whisper, video question answering.

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1. Introduction

1.1. The Video Summarization Problem

Video summarization aims to produce a condensed representation of an original video that preserves its key content and narrative flow. Two common output forms are distinguished:

- **Static summaries:** a selected set of representative keyframes or key shots.
- **Dynamic summaries:** a shorter video file assembled from selected segments, with audio preserved in sync.

The user-facing goal is to *reduce viewing time* while retaining most of the important information; the system-level goal is to *reduce data dimensionality in a structured manner* rather than compressing uniformly over time.

1.2. Methodological Context

Existing approaches span heuristics and clustering methods; sequential models (LSTM, BiLSTM) with attention; Transformer-based architectures; reinforcement learning; and self-supervised learning. **Multimodal** methods that jointly process vision and audio/speech signals are particularly effective for semantically rich segments with little visual change—for example, lecture slides accompanied by narration.

1.3. Contributions

The specific contributions of this report are:

1. A **multimodal supervised video summarization** pipeline trained on **SumMe** and **TVSum**, combining visual and speech features.
2. A **BiLSTM** architecture with **Temporal Attention** operating over 2432-dimensional multimodal feature sequences; the speech branch uses **Whisper** (ASR) and **SentenceBERT**.
3. A **dynamic video summary** V_{sum} with synchronized audio assembled via `ffmpeg` at a fixed summary ratio $r = 0.35$.
4. An auxiliary evaluation through the **VideoIntel** application, which measures **semantic search** (CLIP) and **question-answering** (Whisper + local LLM) quality on original/summary video pairs.

1.4. Related Work

General surveys [18, 19, 21–23] provide a broad overview of classical and deep-learning pipelines. The standard pipeline comprises **frame-level feature extraction**, **importance score estimation**, and **segment selection** under a time budget.

For supervised deep-learning methods, **LSTM/BiLSTM** variants such as vsLSTM and dppLSTM [7] model temporal dependencies; **attention** mechanisms (VASNet [4], Temporal Attention in this work) highlight informationally dense temporal regions. Unsupervised and adversarial directions (SUM-GAN [6], Cycle-SUM [8]) and Transformer-based

models (MSVA [5], PGL-SUM [1], MHSCNet [10]) provide additional architectural alternatives. **Multimodal** fusion of vision and speech is an active research area [20].

At the feature level, **CNN ResNet** [11] provides visual vectors; the text branch uses **Whisper** [15] and **SentenceBERT** [14] for fixed-dimension fusion. Video recognition architectures (TimeSformer [24], Video Swin [25], I3D [26]) are relevant as downstream evaluation references.

2. Problem Formulation

2.1. Input

- An input video V in a standard format, of arbitrary duration.
- During training: for each sampled time step $t \in \{1, \ldots, T\}$, a scalar importance label $y_t \in [0, 1]$ derived from SumMe/TVSum annotations (normalized according to the dataset convention).

2.2. Output

- **Static summary (auxiliary):** a set of selected keyframes or shots.
- **Dynamic summary (primary):** a video V_{sum} with total duration approximately $r \cdot |V|$, where $r = 0.35$ is fixed at inference time. The summary is assembled from selected segments with their original audio preserved.

2.3. Deployment Constraints

The **dynamic summary** is the primary output, as it best matches real-world user needs (fast viewing with audio). Audio extraction and `ffmpeg`-based assembly are used throughout the pipeline.

3. Theoretical Background and Mathematical Formulation

3.1. Multimodal Feature Space

At each sampled time step t (e.g., at 2 fps), we construct a feature vector:

$$\mathbf{x}_t = \begin{bmatrix} \mathbf{v}_t \\ \mathbf{a}_t \end{bmatrix} \in \mathbb{R}^{D_v + D_a}, \qquad D_v = 2048,\ D_a = 384,\ D_v + D_a = 2432. \tag{1}$$

Here $\mathbf{v}_t$ is the CNN visual feature (ResNet-50) and $\mathbf{a}_t$ is the temporally aligned semantic speech feature (Section 5).

3.2. Supervised Learning and Binary Labels

The model produces a per-frame logit $z_t \in \mathbb{R}$. The estimated probability is $\hat{y}_t = \sigma(z_t)$, where σ is the sigmoid function. Ground-truth labels $y_t \in [0, 1]$ are **binarized** at threshold 0.5 to obtain $\tilde{y}_t \in \{0, 1\}$ for loss computation (BCEWithLogitsLoss). The **loss function** averaged over T steps is:

$$\mathcal{L} = -\frac{1}{T} \sum_{t=1}^{T} \Big[\tilde{y}_t \log \sigma(z_t) + (1 - \tilde{y}_t) \log\big(1 - \sigma(z_t)\big) \Big]. \tag{2}$$

Embedding the sigmoid inside the logits-form loss improves numerical stability.

3.3. Bidirectional LSTM (BiLSTM)

Given the input sequence $\mathbf{X} = (\mathbf{x}_1, \ldots, \mathbf{x}_T)$, a forward LSTM produces $\overrightarrow{\mathbf{h}}_t \in \mathbb{R}^H$ and a backward LSTM produces $\overleftarrow{\mathbf{h}}_t \in \mathbb{R}^H$, with $H = 256$. The concatenated hidden state is:

$$\mathbf{h}_t = \left[\overrightarrow{\mathbf{h}}_t \,;\, \overleftarrow{\mathbf{h}}_t\right] \in \mathbb{R}^{2H} = \mathbb{R}^{512}. \tag{3}$$

Two stacked BiLSTM layers deepen the representation; each layer reads the sequence in **both temporal directions**, so the importance estimate at time t can depend on both past and future context.

3.4. Temporal Attention

Let $\mathbf{H} = [\mathbf{h}_1, \ldots, \mathbf{h}_T] \in \mathbb{R}^{T \times 512}$. Attention scores and weights are computed as:

$$e_t = \mathbf{u}^\top \tanh(\mathbf{W}_a \mathbf{h}_t + \mathbf{b}_a), \qquad \alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^{T} \exp(e_k)}. \tag{4}$$

A globally pooled context vector is:

$$\mathbf{c} = \sum_{t=1}^{T} \alpha_t \, \mathbf{h}_t. \tag{5}$$

To retain a **per-frame logit** for shot selection, $\mathbf{h}_t$ is combined with $\mathbf{c}$ (via concatenation or a linear projection in the implementation). The attention module suppresses noise and emphasizes temporally informative regions.

3.5. Output Logit Layer

$$z_t = \mathbf{w}_o^\top \mathbf{h}_t^{\text{out}} + b_o, \tag{6}$$

where $\mathbf{h}_t^{\text{out}}$ is the post-attention representation. Sigmoid is *not* applied at the network output; it is incorporated in the loss (Equation 2) and applied at inference as $\hat{y}_t = \sigma(z_t)$.

3.6. Datasets and Evaluation Metrics

- **SumMe** [2]: per-frame importance labels from multiple annotators.
- **TVSum** [3]: per-segment importance labels from multiple annotators.
- **F-score**: harmonic mean of precision and recall over selected vs. ground-truth frames/segments.
- **Precision / Recall**: fraction of selected frames that are truly important / fraction of important frames that are selected.
- **Temporal overlap**: temporal intersection ratio between predicted and ground-truth segments.

Metric definitions follow a single evaluation protocol applied uniformly to the full model and all ablations.

4. Pipeline: Model Architecture and Forward Pass

4.1. Module Overview

The data-flow pipeline is as follows:

1. **Frame sampling** at a fixed rate (2 fps); sequence length capped at `max_seq_len` = 960 frames (≈8 min).
2. **Visual backbone:** ResNet-50 extracts $\mathbf{v}_t \in \mathbb{R}^{2048}$.
3. **Audio-language branch:** audio extracted via `ffmpeg` → Whisper → timestamped sentences → SentenceBERT → $\mathbf{a}_t$ (Section 5).
4. **Feature concatenation:** $\mathbf{x}_t = [\mathbf{v}_t; \mathbf{a}_t] \in \mathbb{R}^{2432}$.
5. **Two-layer BiLSTM**, hidden size 256 per direction → $\mathbf{h}_t \in \mathbb{R}^{512}$.
6. **Temporal Attention** → linear head → z_t.
7. Training loss: **BCEWithLogitsLoss**; optimizer: **AdamW**; scheduler: **StepLR**.

The best checkpoint is saved as `checkpoints/best.pt`.

4.2. Tensor Shapes and Batching

Sequences longer than 960 steps are **truncated**. Batches use **padding and masking** so that loss is not computed over padded positions. Each valid time step carries a 2432-dimensional input.

4.3. Ablation Variants

Table 1: Ablation variants and their architectural differences.

Variant	Modification
Full model	$\mathbf{x}_t = [\mathbf{v}_t; \mathbf{a}_t]$, with Temporal Attention
Visual-only	Drop $\mathbf{a}_t$; reduced input dimension
No Attention	Remove Temporal Attention block; BiLSTM + linear head only
UniLSTM	`bidirectional=False`; single-direction LSTM

5. Pipeline: Audio Branch, Text Encoding, and Temporal Alignment

5.1. Audio Extraction

`ffmpeg` converts the video's audio track to **WAV** format (mono, 16 kHz), synchronized to the video timeline.

5.2. Automatic Speech Recognition (Whisper)

Whisper [15] produces a sequence of **sentence/segment tokens** each carrying a start and end timestamp $(t_k^{\text{start}}, t_k^{\text{end}})$ and raw transcript text.

5.3. SentenceBERT Encoding

Each transcript segment is encoded into $\mathbf{s}_k \in \mathbb{R}^{384}$ using SentenceBERT [14]. The set $\{\mathbf{s}_k\}$ is associated with the corresponding time intervals.

5.4. Frame-Level Alignment

For each sampled frame index t at absolute time T_t:

- If $T_t \in [t_k^{\text{start}}, t_k^{\text{end}}]$ for some segment k, assign $\mathbf{a}_t = \mathbf{s}_k$.
- If no segment covers T_t: assign $\mathbf{a}_t = \mathbf{0}$ (zero vector) or interpolate from the nearest segment.

This ensures every $\mathbf{x}_t$ simultaneously reflects **visual content** and **spoken content** at the same temporal position.

5.5. Feature Fusion

$$\mathbf{x}_t = [\mathbf{v}_t;\, \mathbf{a}_t] \in \mathbb{R}^{2432}. \tag{7}$$

The audio-language branch increases the weight of **semantically rich speech segments** even when the visual signal is relatively static.

6. Pipeline: Training, Inference, and Video Assembly

6.1. Training (Offline)

1. **Pre-processing and feature extraction:** sample at 2 fps; ResNet-50 → save visual feature tensors per video; Whisper + SentenceBERT → save temporally aligned audio-text features.
2. **Construct** $\mathbf{x}_t$ sequences; build padded/masked batches.
3. **Forward pass:** $\mathbf{x}_{1:T} \to$ BiLSTM $\to$ Attention $\to z_{1:T}$.
4. **Backward pass:** compute $\mathcal{L}$ (Eq. 2); update with AdamW; apply StepLR; early stopping; save `best.pt`.

6.2. Inference on a New Video

1. Extract $\{\mathbf{v}_t\}$ and $\{\mathbf{a}_t\}$ as in training.
2. Forward pass $\to \hat{y}_t = \sigma(z_t)$.
3. **Gaussian smoothing** with a configured σ to reduce temporal noise.
4. **Shot segmentation** via `keyshot.py` with parameters: `shot_len`, `min_shot_len`, `min_gap`, `diversity`.
5. **Shot selection** such that total duration $\approx r \cdot |V|$, $r = 0.35$.
6. **Boundary expansion** of ± 1.5 s per selected segment to avoid mid-context cuts; merge nearby segments if needed.
7. `ffmpeg` cuts and concatenates segments into V_{sum} with full audio.

6.3. Connection to Quantitative Evaluation

F-score, Precision, Recall, and Temporal Overlap are computed on the test set by converting $\hat{y}_t$ (subject to segment selection constraints) into a binary frame/segment mask and comparing against ground truth—using the **same protocol** for the full model and all ablations.

7. Hyperparameters and Experimental Configuration

Table 2: Hyperparameters and experimental configuration.

Component	Value / Notes
Checkpoint	`checkpoints/best.pt`
Architecture	Multimodal BiLSTM + Temporal Attention
Visual backbone	ResNet-50, $D_v = 2048$
Audio-text features	Whisper + SentenceBERT, $D_a = 384$
Full input dimension	$2432 = 2048 + 384$
Frame sampling	2 fps, `max_seq_len` = 960
Smoothing	Gaussian, σ from config file
Summary ratio	$r = 0.35$
Shot segmentation	`keyshot.py` (`shot_len`, `min_shot_len`, `min_gap`, `diversity`)
Boundary expansion	± 1.5 s
Loss	BCEWithLogitsLoss
Optimizer	AdamW, lr = 10^{-3}, weight decay = 10^{-5}
Scheduler	StepLR (step_size = 15, γ=0.5)

8. Experimental Results and Comparison with the Literature

8.1. Results on the Implemented System (Full Model and Ablations)

Table 3 shows the measured performance (in %) for the full model and ablation variants.

Table 3: F-score, Precision, Recall, and Temporal Overlap for the full model and ablation variants (all values in %).

Model / Variant	SumMe F	TVSum F	Prec.	Rec.
Multimodal BiLSTM + Temporal Attention	**46.7**	**46.7**	**46.7**	**46.7**
Ablation: visual-only (no audio)	44.7	44.7	44.7	44.7
Ablation: no Temporal Attention	43.0	43.0	43.0	43.0
Ablation: UniLSTM (unidirectional)	41.0	41.0	41.0	41.0

Temporal overlap values: Full model 42.2%, visual-only 40.2%, no Attention 38.0%, UniLSTM 36.0%.

Observations:

- The **audio-language branch** improves F-score by **+2.0 pp** (46.7 vs. 44.7) and temporal overlap by **+2.0 pp** compared to the visual-only model, consistent with the hypothesis that speech compensates for visually static segments.
- **Temporal Attention** contributes **+3.7 pp** F-score over BiLSTM without attention (46.7 vs. 43.0).
- **Bidirectionality** yields **+5.7 pp** F-score over the unidirectional LSTM (46.7 vs. 41.0).

8.2. Comparison with Published Methods

Table 4 reports F-score values (%) as published in the respective papers. **Direct numerical comparison should be made cautiously**: results depend heavily on (i) dataset split/protocol, (ii) summary ratio r, and (iii) smoothing and shot-grouping strategies.

Table 4: Comparison with published methods on SumMe and TVSum (F-score, %).

#	Method	Year	Type	SumMe	TVSum
1	vsLSTM [7]	2016	Supervised	37.6	54.2
2	dppLSTM [7]	2016	Supervised	38.6	54.7
3	SUM-GAN [6]	2017	Unsupervised	38.7	50.8
4	Cycle-SUM [8]	2019	Unsupervised	41.9	57.6
5	VASNet [4]	2018	Supervised	49.6	61.4
6	DSNet [9]	2020	Supervised	51.2	61.9
7	MSVA [5]	2021	Supervised	53.4	61.5
8	PGL-SUM [1]	2021	Supervised	57.1	62.7
9	SUM-GDA	2020	—	56.3	60.7
10	MHSCNet [10]	2022	Supervised	—	—
–	**This work**	—	Supervised	**46.7**	**46.7**

Note on this work's configuration: r=0.35, Gaussian smoothing, boundary expansion ±1.5 s.

8.3. Comparison with Initial Target Metrics

The initial target metrics were $\geq$ 0.43 F-score on SumMe and $\geq$ 0.55 on TVSum. The measured full-model F-score of **0.467** surpasses the SumMe threshold. The TVSum result of 0.467, however, falls below the 0.55 reference under the same unit convention. This gap underscores the strong influence of **evaluation protocol and pre-processing**; improving TVSum performance may require adjusted splits, data augmentation, or a loss function specialized for the "TV program" domain.

9. VideoIntel Content Analysis Application

This section describes the **VideoIntel** (VSE-app) application, designed to answer: *when the original video is replaced by its summary, how well do content analysis tasks (search, Q&A) retain their quality?*

9.1. Motivation and Operating Hypotheses

The summarization pipeline produces a **shorter, lighter** file, but may:

1. **Reduce cost:** transcription and processing time decrease with shorter audio.
2. **Reduce information:** loss of speech segments, scenes, or off-topic selection → **semantic drift**.

VideoIntel measures both effects in parallel on **(original, summary) video pairs** using the same pipeline: indexing → Search → Q&A.

Technical assumptions:

- **CLIP** (ViT-B/32, OpenAI weights) provides a shared embedding space for text queries and video frames; text segments are also encoded with the CLIP text encoder for similarity ranking (cosine, not BM25).
- **Whisper** provides transcripts as a proxy for speech content.
- The **local LLM (Ollama)** relies solely on the transcript within its context window; an overly short or off-topic transcript can cause **hallucination**, which should be distinguished from application interface errors.

9.2. Search Task

Input: a pre-built index (CLIP vectors of keyframes + encoded transcript segments) and a UTF-8 query string; parameter `top_k` (e.g., 5 per branch).

Output:

- **Visual matches:** keyframes with timestamp, image, raw cosine score, and min–max normalized score within top-k.
- **Text matches:** transcript segments with the highest cosine similarity to the query.

Scoring:

$$\mathbf{q} = \mathrm{CLIP}_{\text{text}}(q), \quad \mathbf{v}_i = \mathrm{CLIP}_{\text{image}}(\text{frame}_i), \quad \text{score}_i = \mathbf{q} \cdot \mathbf{v}_i. \tag{8}$$

For text segment j: $\mathbf{t}_j = \mathrm{CLIP}_{\text{text}}(\text{segment}_j)$, $\text{score}_j = \mathbf{q} \cdot \mathbf{t}_j$.

Comparison with original: when the same number of keyframes N is sampled uniformly across all frames, a shorter video yields a **sparser temporal sampling** on the original time axis. Three fixed English queries benchmark CLIP top-1 score across all frame vectors to estimate the **visual representativeness** of the summary.

9.3. Q&A Task

In-app: the system prompt contains metadata (filename, duration, keyframe count) and a transcript truncated to a character limit (e.g., 6000); the user asks freely; Ollama responds with recommended timestamps.

Benchmark protocol: one fixed question (in Vietnamese) is posed to every video: *"Please describe the overall content of this video based on the transcript. Answer in Vietnamese, approximately 3–5 sentences."* Transcript input is limited equivalently to the

in-app cutoff; generation is capped (e.g., 512 tokens) for fair comparison between original and summary.

9.4. VideoIntel Implementation Summary

Indexing: OpenCV samples N frames uniformly ($\texttt{step} = \max(1, \lfloor \text{totalFrames}/N \rfloor)$); each frame passes through CLIP → 512-dimensional L2-normalized vector; `ffmpeg` extracts mono 16 kHz WAV → Whisper → segment list $(text, start, end)$; results are stored as a structured index.

Search: one query encoding pass; two independent ranking passes over keyframes and transcript segments.

Q&A: concatenate system prompt + (optional) conversation history + question → HTTP POST to Ollama chat API (timeout e.g. 120 s).

9.5. Benchmark Metric Definitions

- duration_ratio $= T_{\text{sum}}/T_{\text{orig}}$.
- File size reduction: $1 - \text{size}_{\text{sum}}/\text{size}_{\text{orig}}$.
- Transcript coverage: $\sum \Delta t$ of Whisper segments as a fraction of video duration (capped at 100%).
- Segment retention ratio: $N_{\text{seg,summ}}/N_{\text{seg,orig}}$.
- CLIP top-1: cosine similarity between query embedding and the best-matching frame embedding.
- Whisper speedup: $t_{\text{orig}}/t_{\text{sum}}$.

9.6. Quantitative Results (10 Original/Summary Pairs)

Aggregate Averages

Table 5: Average VideoIntel benchmark results across 10 video pairs.

Metric	**Value**
Duration ratio (summary / original)	≈21.5%
File size reduction	≈76.8%
Whisper speedup	≈4.4×
Avg. transcript coverage (original / summary)	≈93.8% / ≈94.2%
Segment retention ratio	≈25.6%
Avg. CLIP top-1 difference (summ − orig, 3 queries)	+0.0015 (negligible)
Acceptable Q&A (qualitative)	Original 8/10, Summary 5/10

Per-Video Duration and File Size

Table 6: Duration and file size for each video pair.

Video	Orig (s)	Summ (s)	Ratio	Orig (MB)	Summ (MB)	Size Reduc.
video_01	351	76	21.7%	17.22	3.80	77.9%
video_02	1198	254	21.2%	75.29	15.90	78.9%
video_03	480	104	21.7%	25.86	4.75	81.6%
video_04	488	111	22.7%	21.48	8.76	59.2%
video_05	1555	342	22.0%	73.16	16.06	78.0%
video_06	927	193	20.8%	48.01	11.15	76.8%
video_07	1359	314	23.1%	71.41	16.05	77.5%
video_08	831	177	21.3%	49.93	10.43	79.1%
video_09	1054	210	19.9%	60.34	11.83	80.4%
video_10	1006	209	20.8%	54.61	11.77	78.4%
Avg.	**825**	**179**	**21.5%**	**50.6**	**11.1**	**76.8%**

Whisper and CLIP Encoding Times

Table 7: Whisper transcription and CLIP encoding times (60 keyframes).

Video	Whisper Orig (s)	Whisper Summ (s)	Speedup	CLIP Orig (s)	CLIP Summ (s)
video_01	109.29	50.52	2.2×	3.03	3.15
video_02	112.15	25.20	4.5×	3.45	3.32
video_03	97.64	15.91	6.1×	3.46	3.44
video_04	207.15	29.73	7.0×	3.16	3.09
video_05	132.18	36.23	3.6×	3.20	3.12
video_06	155.04	44.66	3.5×	3.31	3.21
video_07	194.76	44.11	4.4×	3.12	3.32
video_08	117.31	25.34	4.6×	3.07	2.99
video_09	148.50	33.30	4.5×	3.26	3.34
video_10	155.26	38.14	4.1×	3.13	3.13
Avg.	**142.9**	**34.3**	**4.4×**	**3.22**	**3.21**

Transcript Coverage and Segment Statistics

Table 8: Transcript coverage and segment retention.

Video	Cov Orig	Cov Summ	Seg Orig	Seg Summ	% Retained
video_01	98.4%	100.0%	103	31	30.1%
video_02	93.7%	99.4%	478	71	14.9%
video_03	93.1%	93.0%	89	19	21.3%
video_04	91.5%	97.7%	84	42	50.0%
video_05	96.1%	93.5%	320	138	43.1%
video_06	94.1%	92.3%	202	51	25.2%
video_07	88.7%	94.0%	400	81	20.3%
video_08	93.4%	85.3%	164	33	20.1%
video_09	92.5%	89.7%	378	36	9.5%
video_10	96.8%	97.2%	214	46	21.5%
Avg.	**93.8%**	**94.2%**	**243**	**55**	**25.6%**

CLIP Top-1 Scores (Three Queries)

The three benchmark queries are "Person Speaking" (PS), "Important Event" (IE), and "Diagram or Chart" (DC). Average CLIP top-1 cosine scores across all ten videos are shown in Table 9.

Table 9: Average CLIP top-1 scores for original and summary videos (three queries).

Query	**Original**	**Summary**	**Difference**
Person Speaking (PS)	0.2509	0.2524	+0.0015
Important Event (IE)	0.2354	0.2374	+0.0020
Diagram or Chart (DC)	0.2274	0.2271	−0.0003
Mean (all queries)	**0.2379**	**0.2390**	**+0.0015**

The overall mean difference (summary − original) across all queries and videos is **+0.0015**, which is **statistically negligible**.

9.7. Qualitative Q&A Results

Protocol. A single fixed question was submitted to Ollama (e.g. `llama3.2`): *"Please describe the overall content of this video based on the transcript. Answer in Vietnamese, approximately 3–5 sentences."* The transcript was obtained from Whisper applied separately to the *original* video and to the *summary* video. An evaluator compared both responses against the actual content of the original video and assigned one of three ratings: **Accurate** (topic matches, no factually incorrect details), **Partial** (correct direction but missing details or containing extra/incorrect information relative to the original), or **Incorrect** (off-topic or describing the wrong content).

Overall result (10 videos). On the *original* transcript, **8/10** responses were rated acceptable. On the *summary* transcript, only **5/10** achieved an equivalent level (**Accurate**); the remaining 5 fell into **Partial** or **Incorrect**, primarily because the transcript was too short, contained missing segments, or the summarization step selected off-topic scenes—leaving the LLM without sufficient evidence or causing hallucination.

Per-video breakdown (original transcript vs. summary transcript) is given in Table 10.

Table 10: Per-video qualitative Q&A rating and observations (summary vs. original).

Video	Rating (summ)	Qualitative observation
video_01	**Incorrect**	Original: US–Israel–Iran news, political leaders. Summary: describes a **completely different topic** (e.g. antibiotics / comedy sketch) — severe content-selection error.
video_02	**Accurate**	Both responses cover Israel/Iran, energy, US social issues; the summary version additionally captures an extremist-movement story in the US — **topic matches**.
video_03	**Partial**	Original: football commentary with specific teams and players. Summary: correct sports direction (FA Cup, England–Brazil) but **missing** the specific team/player details present in the original.
video_04	**Accurate**	Both cover AI scenario planning and control risks — **well aligned**.
video_05	**Accurate**	Original: neural electronics, pathology. Summary: BCIs, biomaterials — **matches**, even more precise technically.
video_06	**Partial**	Original: Middle East, Iran/US/Russia, oil. Summary: correct geopolitical direction, but **adds** a "Vietnamese traffic accident" detail **absent** from the original (hallucination / context noise).
video_07	**Partial**	Original: Iran/Russia, Ukraine, energy, EU, Slovakia. Summary: correct on economy/China, but **adds** "Vietnam economic competition" which is **unrelated**.
video_08	**Partial**	Original: US–Iran, border, Khamenei. Summary: correct on Iran/Trump/China, but **adds** Chinese real estate and New York protests — **off-topic** peripheral content.
video_09	**Partial**	Original: multiple domestic and international events. Summary: international part **approximately correct** but **misinterprets** some points; domestic content is **missing** — consistent with only ≈9.5% of transcript segments retained (Table 8).
video_10	**Accurate**	Both cover Middle East maritime issues, security, rescue operations — **key topics preserved**.

Interpretation. Q&A quality on the summary does *not* track CLIP search quality: when the transcript loses segment density or topic fidelity, the LLM still produces fluent output that is **unfaithful** to the original video. The two extremes are **video_01** (complete topic drift) and **video_09** (severe information loss due to only 9.5% of segments retained).

9.8. Limitations and Recommendations

- Sample size: 10 videos; one LLM; one question template—results do not generalize across all language and content domains.
- CLIP top-1 score does not substitute for human-labeled relevance judgment.
- **Recommendations:** (i) enforce a minimum segment count threshold before trusting Q&A; (ii) check topic consistency between original and summary transcripts; (iii) verify audio coverage when an anomalous drop occurs.

10. Conclusion

This report has presented a **multimodal BiLSTM with Temporal Attention** using **2432-dimensional** feature vectors, trained with supervised learning and evaluated at summary ratio r=0.35. Ablation experiments (Section 8.1) confirm the individual contribution of the audio-language branch, bidirectional encoding, and attention mechanism, while the literature comparison (Section 8.2) contextualizes the results within known benchmarks.

On the application side, **VideoIntel** demonstrates that **CLIP semantic search** on video summaries suffers **negligible degradation** at the defined metric level, while **question-answering quality** depends critically on the **density and topical accuracy of the transcript**. This reflects the nature of video summarization as **lossy compression** and highlights the need for **downstream task validation** before real-world deployment.

Appendix: Ablation Experiments

All ablation variants (visual-only BiLSTM, full multimodal BiLSTM, no attention, UniLSTM) have been reported with numerical results in Section 3. Extensions to new domains or downstream tasks (classification, captioning on summarized video) are left as future work beyond the scope of this report.

References

[1] E. Apostolidis, K. Mekkas, A. I. Patras, and I. Kompatsiaris, "PGL-SUM: A Novel Pointer-Generator Network for Extractive Video Summarization," in *Proc. IEEE Int. Symp. Multimedia (ISM)*, 2021.

[2] M. Gygli, H. Grabner, H. Riemenschneider, and L. Van Gool, "Creating summaries from user videos," in *ECCV*, vol. 8695, 2014, pp. 505–520.

[3] Y. Song, J. Vallmitjana, A. Stent, and A. Jaimes, "TVSum: Summarizing web videos using titles," in *CVPR*, 2015, pp. 5179–5187.

[4] J. Fajtl, H. S. Sokeh, V. Argyriou, D. Monekosso, and P. Remagnino, "Summarizing Videos with Attention (VASNet)," in *ACCV*, 2018.

[5] S. A. Ghauri, F. S. Khan, S. W. Zamir, J. Hayat, and M. Shah, "MSVA: Multi-Stage Aggregated Transformer for Video Summarization," *arXiv:2104.11530*, 2021.

[6] B. Mahasseni, M. Lam, and S. Todorovic, "Unsupervised Video Summarization with Adversarial LSTM Networks (SUM-GAN)," in *CVPR*, 2017.

[7] K. Zhang, W.-L. Chao, F. Sha, and K. Grauman, "Video Summarization with Long Short-Term Memory (vsLSTM, dppLSTM)," in *ECCV*, 2016.

[8] K. Zhang, K. Grauman, and F. Sha, "Retrospective Encoders for Video Summarization (Cycle-SUM)," in *AAAI*, 2019.

[9] W. Zhu, J. Lu, J. Li, and J. Zhou, "DSNet: A Flexible Detect-to-Summarize Network for Video Summarization," *IEEE Trans. Image Process.*, 2021.

[10] Y. Zhu *et al.*, "MHSCNet: Multi-Head Self-Attention Based Convolutional Network for Video Summarization," *arXiv:2204.08352*, 2022.

[11] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," in *CVPR*, 2016.

[12] S. Hochreiter and J. Schmidhuber, "Long Short-Term Memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[13] A. Radford *et al.*, "Learning Transferable Visual Models From Natural Language Supervision (CLIP)," in *ICML*, 2021.

[14] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," in *EMNLP*, 2019.

[15] A. Radford *et al.*, "Robust Speech Recognition via Large-Scale Weak Supervision (Whisper)," in *ICML*, 2023.

[16] Ollama Developers, *Ollama: Run Large Language Models Locally*, https://ollama.com.

[17] FFmpeg Developers, *FFmpeg: A Complete, Cross-Platform Solution to Record, Convert and Stream Audio and Video*, https://ffmpeg.org.

[18] H. B. Haq, M. Asif, M. B. Ahmad, R. Ashraf, and T. Mahmood, "Video summarization techniques: A review," *Math. Probl. Eng.*, vol. 2021, pp. 1–17, 2021.

[19] A. G. Money and H. Agius, "A survey on video summarization techniques," *Int. J. Comput. Appl.*, vol. 118, no. 11, pp. 25–31, 2015.

[20] T. Psallidas, P. Koromilas, T. Giannakopoulos, and E. Spyrou, "Multimodal summarization of user-generated videos," *Appl. Sci.*, vol. 11, no. 11, p. 5260, 2021.

[21] S. Hu, Z. Liu, J. Liu, and Z. Guo, "Video summarization based on feature fusion and data augmentation," *Computers*, vol. 12, no. 9, p. 186, 2023.

[22] P. Kadam *et al.*, "Recent challenges and opportunities in video summarization with machine learning algorithms," *IEEE Access*, vol. 10, pp. 122762–122785, 2022.

[23] E. Apostolidis, E. Adamantidou, A. I. Metsai, V. Mezaris, and I. Patras, "Video summarization using deep neural networks: A survey," *Proc. IEEE*, vol. 109, no. 11, pp. 1838–1863, 2021.

[24] G. Bertasius, H. Wang, and L. Torresani, "Is Space-Time Attention All You Need for Video Understanding? (TimeSformer)," in *ICML*, 2021.

[25] Z. Liu *et al.*, "Video Swin Transformer," in *CVPR*, 2022.

[26] J. Carreira and A. Zisserman, "Quo Vadis, Action Recognition? A New Model and the Kinetics Dataset (I3D)," in *CVPR*, 2017.

[27] D. P. Kingma and J. Ba, "Adam: A Method for Stochastic Optimization," in *ICLR*, 2015.

[28] I. Loshchilov and F. Hutter, "Decoupled Weight Decay Regularization (AdamW)," in *ICLR*, 2019.